

12.3 Tree-decompositions

Trees are graphs with some very distinctive and fundamental properties; consider Theorem 1.5.1, or the more sophisticated example of Kruskal's theorem. It is therefore legitimate to ask to what degree those properties can be transferred to more general graphs, graphs that are not themselves trees but tree-like in some sense.² In this section, we study a concept of tree-likeness that permits generalizations of all the tree properties referred to above (including Kruskal's theorem), and which plays a crucial role in the proof of the graph minor theorem.

Let G be a graph, T a tree, and let $\mathcal{V} = (V_t)_{t \in T}$ be a family of bags $V_t \subseteq V(G)$ indexed by the nodes t of T . The pair $(T, \mathcal{V})$ is called a *tree-decomposition* of G if it satisfies the following three conditions:

- (T1) $V(G) = \bigcup_{t \in T} V_t$;
- (T2) for every edge $e \in G$ there exists a $t \in T$ such that both ends of e lie in V_t ;
- (T3) $V_{t_1} \cap V_{t_3} \subseteq V_{t_2}$ whenever $t_1, t_2, t_3 \in T$ satisfy $t_2 \in t_1 T t_3$.

Conditions (T1) and (T2) together say that G is the union of the subgraphs $G[V_t]$. We call these subgraphs the *parts* of $(T, \mathcal{V})$, and say that $(T, \mathcal{V})$ is a tree-decomposition of G into the bags V_t or the parts $G[V_t]$. Condition (T3) implies that the bags of $(T, \mathcal{V})$ are organized roughly like a tree (Fig. 12.3.1).

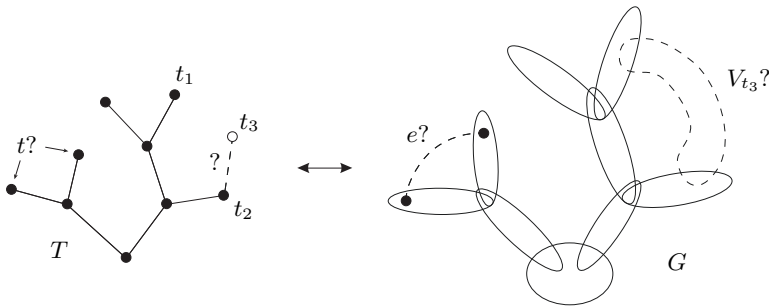


Fig. 12.3.1. Edges and bags ruled out by (T2) and (T3)

Before we discuss the role that tree-decompositions play in the proof of the minor theorem, let us note some of their basic properties. Consider a fixed tree-decomposition $(T, \mathcal{V})$ of G , with $\mathcal{V} = (V_t)_{t \in T}$ as above.

Perhaps the most important feature of a tree-decomposition is that it transfers the separation properties of its tree to the graph decomposed:

² What exactly this 'sense' should be will depend both on the property considered and on its intended application.

[12.4.3]
[12.5]
[12.6.5]

Lemma 12.3.1. Let t_1t_2 be any edge of T and let T_1, T_2 be the components of $T - t_1t_2$, with $t_1 \in T_1$ and $t_2 \in T_2$. Then $V_{t_1} \cap V_{t_2}$ separates $U_1 := \bigcup_{t \in T_1} V_t$ from $U_2 := \bigcup_{t \in T_2} V_t$ in G (Fig. 12.3.2).

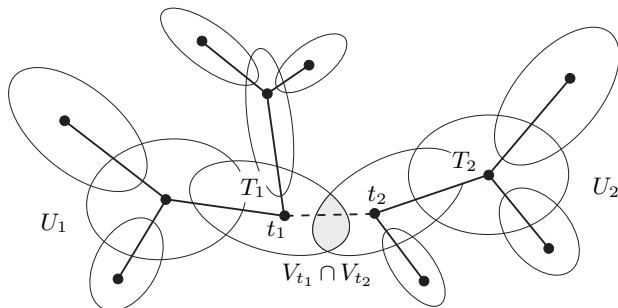


Fig. 12.3.2. $V_{t_1} \cap V_{t_2}$ separates U_1 from U_2 in G

Proof. Consider a shortest U_1 – U_2 path. By (T1) it has length at most 1, so by (T2) it has length 0. By (T3), its vertex lies in $V_{t_1} \cap V_{t_2}$. $\square$

induced
separation

We shall say that the separation $\{U_1, U_2\}$ of G in Lemma 12.3.1 is *induced* by the edge t_1t_2 of T , or more generally by $(T, \mathcal{V})$. Its separator $U_1 \cap U_2 = V_{t_1} \cap V_{t_2}$ is the *adhesion set* of V_{t_1} and V_{t_2} .

adhesion
torsos

The *adhesion* of a tree-decomposition is the maximum size of its adhesion sets. (If T is trivial, we let it be zero.) The *torsos* of a tree-decomposition are the supergraphs of its parts $G[V_t]$ obtained by making their adhesion sets complete: by adding to $G[V_t]$ any edges not in G whose ends lie in a common adhesion set $V_t \cap V_{t'}$ with $tt' \in E(T)$.

Tree-decompositions are passed on to subgraphs, indeed to minors:

[12.4.1]
[12.4.4]
[12.4.3]
[12.6.2]

Lemma 12.3.2. For every $H \subseteq G$, the pair $(T, (V_t \cap V(H))_{t \in T})$ is a tree-decomposition of H . $\square$

[12.4.1]
[12.4.3]

Lemma 12.3.3. Suppose that $G = IH$ with branch sets U_h , $h \in V(H)$. Let $f: V(G) \rightarrow V(H)$ be the map assigning to each vertex of G the index of the branch set containing it. For each $t \in T$ let $W_t := \{f(v) \mid v \in V_t\}$, and put $\mathcal{W} := (W_t)_{t \in T}$. Then $(T, \mathcal{W})$ is a tree-decomposition of H .

Proof. The assertions (T1) and (T2) for $(T, \mathcal{W})$ follow immediately from the corresponding assertions for $(T, \mathcal{V})$. Now let $t_1, t_2, t_3 \in T$ be as in (T3), and consider a vertex $h \in W_{t_1} \cap W_{t_3}$ of H ; we show that $h \in W_{t_2}$. By definition of W_{t_1} and W_{t_3} , the set U_h meets both V_{t_1} and V_{t_3} . As U_h is connected in G , this implies by Lemma 12.3.1 that U_h also meets V_{t_2} . Hence $h \in W_{t_2}$, by definition of W_{t_2} . $\square$

Here is another useful consequence of Lemma 12.3.1:

Lemma 12.3.4. *Any set of vertices not contained in a bag of $(T, \mathcal{V})$ contains two vertices that are separated by an adhesion set of $(T, \mathcal{V})$.* [12.4.3]

Proof. Given $W \subseteq V(G)$, orient the edges of T as follows. For each edge $t_1 t_2 \in T$, define U_1, U_2 as in Lemma 12.3.1; then $V_{t_1} \cap V_{t_2}$ separates U_1 from U_2 . Unless $V_{t_1} \cap V_{t_2}$ separates two vertices from W , we can find an $i \in \{1, 2\}$ such that $W \subseteq U_i$, and orient $t_1 t_2$ towards t_i .

Let t be the last node of a maximal directed path in T ; then all the edges of T at t are oriented towards t . We claim that $W \subseteq V_t$. Given $w \in W$, let $t' \in T$ be such that $w \in V_{t'}$. If $t' \neq t$, then the edge e at t that separates t' from t in T is directed towards t , so w also lies in $V_{t''}$ for some t'' in the component of $T - e$ containing t . Therefore $w \in V_t$ by (T3). $\square$

The following special case of Lemma 12.3.4 is used particularly often:

Corollary 12.3.5. *Every complete subgraph of G is contained in some part of $(T, \mathcal{V})$.* $\square$ [12.6.2]

The tree-decomposition $(T, \mathcal{V})$ of G is called *simplicial* if all the separators $V_{t_1} \cap V_{t_2}$ induce complete subgraphs in G . This assumption can enable us to lift assertions about the parts of the decomposition to G itself. For example, if all the parts in a simplicial tree-decomposition of G are k -colourable, then so is G (Exercise 19). The same applies to the property of not containing a K^r minor for some fixed r .

simplicial

Conversely, if G can be constructed recursively from a set $\mathcal{H}$ of graphs by pasting along complete subgraphs, then G has a simplicial tree-decomposition into elements of $\mathcal{H}$. For example, by Wagner's Theorem 7.3.4, any graph without a K^5 minor has a supergraph with a simplicial tree-decomposition into plane triangulations and copies of the Wagner graph W , and similarly for graphs without K^4 minors (see Proposition 12.6.2).

Tree-decompositions may thus lead to intuitive structural characterizations of graph properties. A particularly simple example is the following characterization of chordal graphs:

Proposition 12.3.6. *G is chordal if and only if G has a tree-decomposition into complete parts.* [12.4.4]
[12.6.2]

Proof. We apply induction on $|G|$. We first assume that G has a tree-decomposition $(T, \mathcal{V})$ such that $G[V_t]$ is complete for every $t \in T$; let us choose $(T, \mathcal{V})$ with $|T|$ minimum. If $|T| \leq 1$, then G is complete and hence chordal. So let $t_1 t_2 \in T$ be an edge, and for $i = 1, 2$ define T_i and $G_i := G[U_i]$ as in Lemma 12.3.1. Then $G = G_1 \cup G_2$ by (T1) and (T2), and $V(G_1 \cap G_2) = V_{t_1} \cap V_{t_2}$ by the lemma; thus, $G_1 \cap G_2$ is complete. Since $(T_i, (V_t)_{t \in T_i})$ is a tree-decomposition of G_i into complete (5.5.1)

parts, both G_i are chordal by the induction hypothesis. (By the choice of $(T, \mathcal{V})$, neither G_i is a subgraph of $G[V_{t_1} \cap V_{t_2}] = G_1 \cap G_2$, so both G_i are indeed smaller than G .) Since $G_1 \cap G_2$ is complete, any induced cycle in G lies in G_1 or in G_2 and hence has a chord, so G too is chordal.

Conversely, assume that G is chordal. If G is complete, there is nothing to show. If not then, by Proposition 5.5.1, G is the union of smaller chordal graphs G_1, G_2 with $G_1 \cap G_2$ complete. By the induction hypothesis, G_1 and G_2 have tree-decompositions $(T_1, \mathcal{V}_1)$ and $(T_2, \mathcal{V}_2)$ into complete parts. By Corollary 12.3.5, $G_1 \cap G_2$ lies inside one of those parts in each case, say with indices $t_1 \in T_1$ and $t_2 \in T_2$. As one easily checks, $((T_1 \cup T_2) + t_1 t_2, \mathcal{V}_1 \cup \mathcal{V}_2)$ is a tree-decomposition of G into complete parts. $\square$

Let us wind up this section with an application of tree-decompositions to connectivity that generalizes the idea of the block-cutvertex tree from Lemma 3.1.4. A set $U \subseteq V(G)$ of at least k vertices is $(< k)$ -inseparable in G if no two vertices from U can be separated in G by fewer than k other vertices (which may or may not lie in U).

k-block

A maximal $(< k)$ -inseparable set of vertices is a *k*-block. Thus, the 1-blocks of a graph are its components; its 2-blocks are the non-singleton vertex sets spanning a block as defined in Chapter 3. In general, a *k*-block need not induce a highly connected subgraph: the many paths between its vertices that are needed to make it $(< k)$ -inseparable can all lie outside it. Its ‘connectivity’ is thus measured in the ambient graph; its vertices themselves may even be independent.

Theorem 12.3.7. *For every integer $k \geq 1$, every graph G has a tree-decomposition $(T, \mathcal{V})$ with the following properties:*

- (i) $(T, \mathcal{V})$ has *adhesion* $< k$.
- (ii) *Distinct k -blocks lie in different bags. Moreover, every two blocks are separated by an adhesion set that is no larger than the smallest set of vertices that separates them in G .*
- (iii) *Every automorphism of G acts on the set of bags of $(T, \mathcal{V})$, and the action on $V(T)$ which this induces is an automorphism of T .*

Note that, by (i) and Lemma 12.3.4, every *k*-block is contained in some bag. This is unique by (i) and (T3), so the bags in (ii) are well defined. Assertion (iii) means that, for every automorphism φ of G and every $t \in T$, the set $\varphi(V_t)$ is another bag $V_{t'}$ (possibly V_t), and this map $\varphi: t \mapsto t'$ is an automorphism of T .³ Such tree-decompositions are called *canonical*.

³ This can be important both theoretically and in applications. For example, when a computer constructs a tree-decomposition, then this may depend on how we present the graph to it as input. But if it finds a canonical decomposition, then this will always be the same up to automorphism, in the sense just defined.

12.4 Tree-width

As indicated by Figure 12.3.1, the bags of $(T, \mathcal{V})$ reflect the structure of the tree T , so in this sense the graph G decomposed resembles a tree. However, this is valuable only inasmuch as the structure of G within each part is negligible: the smaller the bags, the closer the resemblance.

This observation motivates the following definition. The *width* of $(T, \mathcal{V})$ is the number

$$\max \{ |V_t| - 1 : t \in T \},$$

and the *tree-width* $\text{tw}(G)$ of G is the least width of any tree-decomposition of G . As one easily checks,⁴ trees themselves have tree-width 1.

By Lemmas 12.3.2 and 12.3.3, the tree-width of a graph will never be increased by deletion or contraction:

Lemma 12.4.1. *If $H \preceq G$ then $\text{tw}(H) \leq \text{tw}(G)$.* □

Graphs of bounded tree-width are sufficiently similar to trees that it becomes possible to adapt the proof of Kruskal's theorem to the class of these graphs; very roughly, one has to iterate the 'minimal bad sequence' argument from the proof of Lemma 12.1.3 $\text{tw}(G)$ times. This takes us a step further towards a proof of the graph minor theorem:

Theorem 12.4.2. (Robertson & Seymour 1990)

For every integer $k > 0$, the graphs of tree-width $< k$ are well-quasi-ordered by the minor relation.

In order to make use of Theorem 12.4.2 for a proof of the full graph minor theorem, we should be able to say something about the graphs which it does not cover, i.e., to deduce some information about a graph from the assumption that its tree-width is large. Our next result, the *tree-width duality theorem*, achieves just that: it identifies a canonical obstruction to small tree-width, a structural phenomenon that occurs in a graph if and only if its tree-width is large.

Let us say that two subsets of $V(G)$ *touch* if they have a vertex in common or G contains an edge between them. A set of mutually touching connected vertex sets in G is a *bramble*. Extending our terminology of Chapter 2, we say that a subset of $V(G)$ *covers* (or is a *cover* of) a bramble $\mathcal{B}$ if it meets every element of $\mathcal{B}$. The least number of vertices covering a bramble is the *order* of that bramble. A *k-bramble* is one of order k .

⁴ Indeed the '−1' in the definition of width serves no other purpose than to make this statement true.